DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
UNIVERSITY OF BARISHAL
Course Title: Data Structures
Course Code: CSE-1201
1st Year 2nd Semester
Session: 2019-20

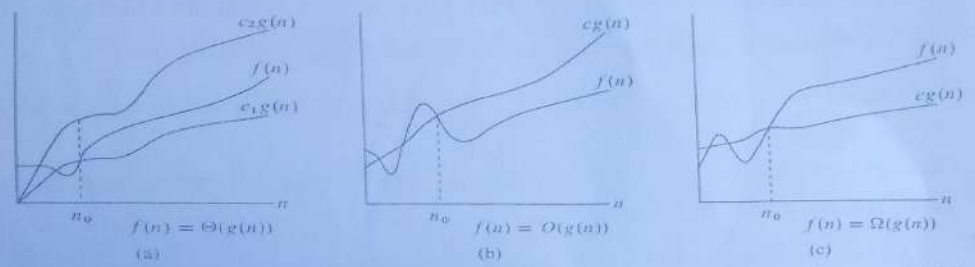
Time: 3 Hours **Marks:** 60

(Answer any FIVE questions)

1. a) Differentiate between data types and data structure. Why should you have good knowledge on data structures as CSE student? [3]

   b) Write down the algorithm for Insertion sort and analyze the best case and worst case running time for the algorithm. [3]

   c) What do you mean by growth of function? Briefly describe the following (a), (b) and (c) of the figure. [3]



   d) Write down the best case, average case and worst case complexity for the Insertion sort, Quick sort, Merge sort, Heap sort and Bubble sort. [3]

2. a) Sort the items using Shell Sort algorithm: 7, 19, 24, 13, 31, 8, 82, 18, 44, 63, 5, and 29. [3]

   b) Quicksort can be described as a recursive in-place sorting algorithm that performs a partition() operation on the given array and then invokes itself twice on two distinct sub ranges of the array. Now, [5]

      i. Sort the following array A=<2, 8, 7, 1, 3, 5, 6, 4> using the quick sort algorithm.
      ii. Describe the purpose, I/O parameters and effect of the partition() procedure and explain what the pivot is.
      iii. Give pseudocode for the quicksort() procedure that would call the partition() procedure you described earlier.
      iv. Some researchers have suggested choosing the pivot from a randomly chosen location in the input array. Discuss the advantages and disadvantages of such a solution. How does it affect the worst-case and average-case behavior?

   c) Consider the following heap: [4]

      For each of the following questions use the original heap. Do not use the heap obtained after each answer.

      i) Represent the heap as an array indexed by 1 through 9.
      ii) Delete the element 86 from the heap shown in the above figure.
      iii) Delete the element 11 from the heap shown in the above figure.
      iv) Insert the element 25 to the heap.
      v) Is it max heap or min heap? Sorting the elements in descending order by heap sort.

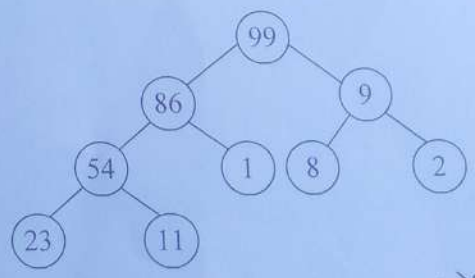


Figure for Q-2(c)

3. a) Define divide and conquer. A well-known problem in UVA- 374: Big Mod Problem, calculate $B^P \% M$ for large B, P and M using an efficient algorithm. Write down the program for calculating the solutions. [3]

   b) What is the data structures used to perform recursion? Write down the recursive formula for the following problems: i) Fibonacci Number and ii) GCD iii) Tower of Hanoi [3]

   c) Why do we need postfix and prefix notation over infix notation? Translate the infix notation $A*(B+D)/E-F*(G+H/K)$ to its equivalent postfix notation using stack. [4]

   d) What are the differences between heap and BST? When to use a heap and when to use a BST? [2]

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4. a) Show how to implement a queue using two stacks. Analyze the running time of the queue operations. [2]

b) Consider the following stack of characters, where STACK is allocated N=8 memory cells. [3]
STACK: A, C, D, F, K, _, _, _. (_ means empty allocated cell)
Describe the stack as the following operations takes place:

i. POP(STACK, ITEM)
ii. POP(STACK, ITEM)
iii. POP(STACK, ITEM)
iv. PUSH(STACK, R)
v. PUSH(STACK, L)
vi. PUSH(STACK, S)
vii. PUSH(STACK, P)
viii. POP(STACK, ITEM)

c) Consider the problem in **4. b)**. (i) When will overflow occur? (ii) When will C be deleted before R? [2]

d) Draw a picture of the sequence {13, 4, 8, 19, 5, 11} stored as a doubly linked list using the multiple-array representation. Do the same for the single-array representation. [5]

5. a) Write down the program to insert and delete operations in double linked list considering all the cases. [4]

b) Consider a hash table of size 7 with hash function $h(k) = k \mod 7$. Draw the table that results after inserting in the given order, the values, 19, 26, 13, 48, 17 for each of the scenarios below: [4]

i) When collisions are handled by separate chaining.
ii) When collisions are handled by linear probing.
iii) When collisions are handled by double hashing.

Using a second hash function $h'(k) = 5 - (k \mod 5)$.

c) Suppose the following six numbers are inserted into an empty binary search tree: 33, 50, 45, 52, 12, 10. Show the tree as each number is inserted into a binary search tree. [3]

d) How can you differentiate hash table from the direct addressing table? [1]

6. a) **What do you mean by disjoint set data structure?** Suppose that CONNECTED-COMPONENTS is run on the undirected graph G=(V, E), where V = { a, b, c, d, e, f, g, h, i, j, k} and the edges of E are processed in the order (d, i), (f, k), (g, i), (b, g), (a, h), (i, j), (d, k), (b, j), (d, f), (g, j), (a, e). **List the vertices in each connected component after each iteration of lines 3–5.** [3]

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CONNECTED-COMPONENTS(G)
1  for each vertex v ∈ G.V
2      MAKE-SET(v)
3  for each edge (u, v) ∈ G.E
4      if FIND-SET(u) ≠ FIND-SET(v)
5          UNION(u, v)
```

b) Consider the following AVL tree: [4]

i) Insert the key 30 into the tree and rebalance if needed. Draw the final and all intermediate trees that you need.
ii) Remove the key 40 from the original tree and rebalance if needed. Draw the final tree and all intermediate trees that you need.

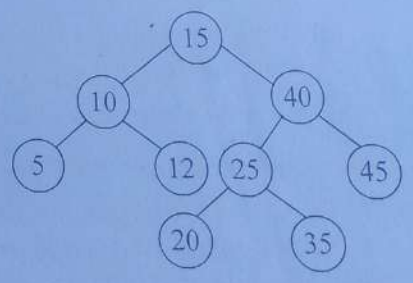

Figure for Q-6(b)

c) Consider the following queue of characters, where QUEUE is a circular array which is allocated six memory cells: [3]

FRONT=2, REAR=4 and QUEUE: _, A, C, D, _, _.

Describe the queue as the following operations take place:

i) F is added to the queue. ii) Two letters are deleted.
iii) K, L and M are added to the queue. iv) Two letters are deleted.

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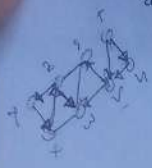
d) Define priority queue. Show how to implement a first-in, first-out queue with a priority queue. Show how to implement a stack with a priority queue. [2]

7. a) Consider the following adjacent list for directed graph. [5]
adj(y) = [x], adj(x) = [z], adj(z) = [y,w], adj(w) = [x], adj(s) = [z,w], adj(v) = [s,w], adj(t) = [u,v], adj(u) = [v].



i. Draw a graph from the above information
ii. Draw the adjacency and incidence matrix for the above graph. What are the advantages of adjacency matrix over adjacency list for graph traversal?
iii. Identify the tree edge, back edge, cross edge and forward edge for the graph.

b) Write down the algorithm for the Breadth First Search with suitable illustration. [4]

c) What do you mean by MST? Find the MST for the following graph using PRIM'S algorithm. Is prims a Greedy algorithm? [3]

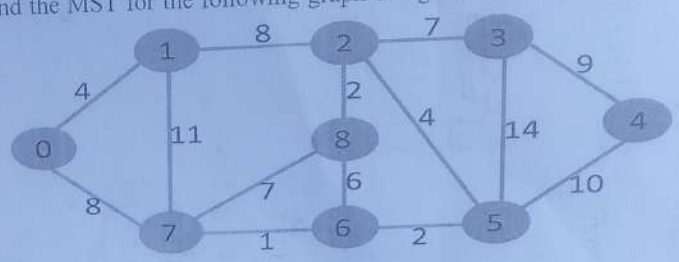


Figure for Q-7(c)

8. a) Prove that Figure Q-8(a)(ii) can be derived from Q-8(a)(i) by using Strongly Connected Components(SCC) algorithms. Justify your answer. [4]

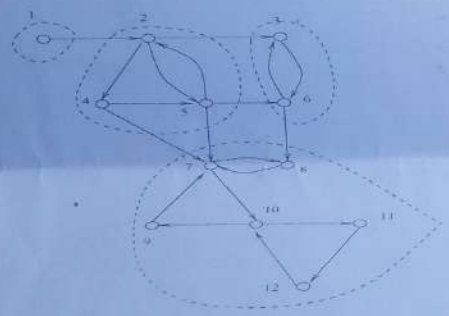


Figure for Q-8(a)(i)

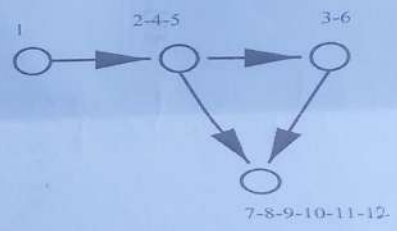


Figure for Q-8(a)(ii)

b) Define complete binary tree. Traverse the given tree using In-order, Preorder and Post-order traversals. [2]

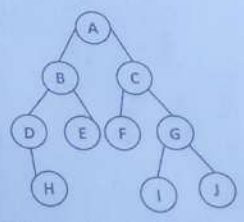
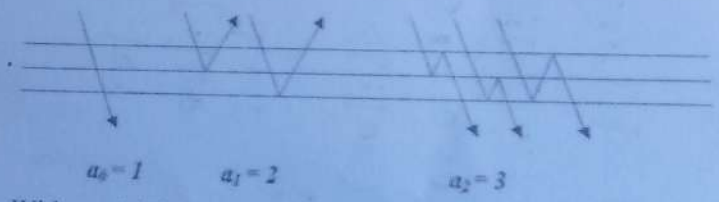


Figure for Q-8(b)

c) A well renowned UVA Problem: 10334, Suppose we put two panes of glass back-to-back. How many ways a_n are there for light rays to pass through or be reflected after changing direction n times? Following figure shows the situations when the value of n is 0, 1 and 2. Now your task is to write down the program for input 0, 1, 2: output will be 1, 2, and 3 respectively. [3]



d) Working Modulo With $q = 11$, how many spurious hits does Rabin-Karp encounter when looking for pattern, $P = 26$ in text, $T = 3141592653589793$? Elaborate your answer. [3]

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